

AI Academic Operations Platform (V1)

Detailed Project Blueprint

Project Vision

- Build an AI-first Academic Operations Platform rather than a simple timetable application.
- Focus on one domain extremely well: academic scheduling and faculty operations.
- Architecture should be modular so future university ERP modules can be added.

Core Modules

- Authentication & Role-Based Access Control
- Faculty Management
- Leave Management
- Dynamic Timetable Engine
- Classroom & Seating Management
- AI Platform
- Analytics Dashboard
- Notification Center

Faculty Management

- Faculty profiles
- Departments
- Subjects & expertise
- Availability matrix
- Weekly workload
- Maximum workload
- Preferred timings
- Preferred classrooms
- Office hours
- Leave balance
- Alternate faculty
- Employment details
- Calendar integration

Leave Management

- Apply leave
- Leave approval workflow
- Leave calendar
- Leave balance

- Conflict detection
- Automatic substitute suggestions
- Timetable impact analysis
- Approval history
- Audit logs

Dynamic Timetable Engine

- AI timetable generation
- Manual timetable editor
- Weekly/Daily/Semester views
- Faculty timetable
- Student timetable
- Room timetable
- Laboratory scheduling
- Conflict detection
- Automatic optimization
- Version history
- Dynamic rescheduling
- Holiday adjustments
- Substitute timetable
- What-if simulation
- Explainable AI recommendations

Classroom & Seating Management

- Building management
- Room management
- Lab management
- Capacity management
- Equipment tracking
- Maintenance status
- Regular seating
- Exam seating
- Random seating
- Hall allocation
- Invigilator allocation
- Accessibility support

AI Platform

- LLM Gateway

- Prompt Library
- Memory
- RAG
- Embeddings
- Vector Database
- LangChain
- LangGraph
- Coordinator Agent
- Faculty Agent
- Leave Agent
- Timetable Agent
- Room Agent
- Conflict Agent
- Notification Agent
- Approval Agent
- Analytics Agent
- Guardrails
- Evaluation
- LLMOps

Knowledge Base (RAG)

- University regulations
- Academic calendar
- Faculty handbook
- Leave policies
- Timetable rules
- Circulars
- Department guidelines
- Holiday notifications

Dashboards

- Admin Dashboard
- HOD Dashboard
- Faculty Dashboard
- Student Dashboard
- AI Decision Center

Analytics

- Faculty utilization

- Room utilization
- Leave analytics
- Conflict statistics
- Timetable efficiency
- AI recommendations
- Operational reports

Technology Stack

- Frontend: React + TypeScript + Tailwind CSS
- Backend: FastAPI
- Database: PostgreSQL
- Cache: Redis
- Vector Store: ChromaDB or pgvector
- AI: OpenAI-compatible LLM + LangGraph + LangChain
- Docker
- AWS
- CI/CD
- Monitoring & Logging

Version 1 Scope

- Authentication
- Faculty Management
- Leave Management
- Dynamic Timetable
- Classroom Management
- Seating Arrangement
- AI Assistant
- RAG
- Multi-Agent Workflows
- Analytics
- Notifications

Future Versions (Not in V1)

- Admissions
- Finance
- Hostel
- Transport
- Library
- Placements

- HR
- Research
- IoT
- Digital Twin
- Predictive ML
- Knowledge Graph